13.12.1 Stage: I Mesophyll Cells

Phosphoenol Pyruvate + CO_2
(PEP) (3C)

PEP carboxylase

Oxaloacetic acid (OAA) (4C)

Oxaloacetic acid (OAA) is converted into malic acid or aspartic acid and is transported to the bundle sheath cells through plasmodesmata.

13.12.2 Stage: II Bundle Sheath Cells

Malic acid undergoes decarboxylation and produces a 3 carbon compound Pyruvic acid and CO_2. The released CO_2 combines with RUBP and follows the calvin cycle and finally sugar is released to the phloem. Pyruvic acid is transported to the mesophyll cells.

$$\underset{(5C)}{RUBP} + CO_2 \xrightarrow{\text{Rubisco}} \underset{(3C)}{2\ PGA}$$

Activity

- Collect the leaves of Paddy (C_3) and Sugar cane (C_4).
- Take the cross section.
- Observe the sections under the microscope.
- See the difference in their anatomy (Dimorphic chloroplast and Kranz anatomy).



Kranz Anatomy: It is the German term meaning a halo or wreath. In C_4 plants vascular bundles are surrounded by a layer of bundle sheath. Bundle sheath is surrounded by a ring of mesophyll cells. The characteristic feature of C_4 plants is the presence of dimorphic chloroplast:

Bundle sheath chloroplast: Larger chloroplast, thylakoids not arranged in granum and rich in starch.

Mesophyll Chloroplast: Smaller chloroplast, thylakoids arranged in granum and less starch.

13.12.3 Significance of C_4 cycle

1. Plants having C_4 cycle are mainly of tropical and sub-tropical regions and are able to survive in environment with low CO_2 concentration.
2. C_4 plants are partially adapted to drought conditions.
3. Oxygen has no inhibitory effect on C_4 cycle since PEP carboxylase is insensitive to O_2.
4. Due to absence of photorespiration, CO_2 Compensation Point for C_4 is lower than that of C_3 plants.

Differences between C_3 Plants (C_3 Cycle) and C_4 Plants (C_4 Cycle) are given in table 13.4.

Table 13.4: Differences between C_3 and C_4 plants

C_3 Plants	C_4 Plants
1. CO_2 fixation takes place in mesophyll cells only	1. CO_2 fixation takes place mesophyll and bundle sheath
2. CO_2 acceptor is RUBP only	2. PEP in mesophyll and RUBP in bundle sheath cells
3. First product is 3C- PGA	3. First product is 4C- OAA
4. Kranz anatomy is not present	4. Kranz anatomy is present
5. Granum is present in mesophyll cells	5. Granum present in mesophyll cells and absent in bundle sheath
6. Normal Chloroplast	6. Dimorphic chloroplast
7. Optimum temperature 20° to 25°C	7. Optimum temperature 30° to 45°C
8. Fixation of CO_2 at 50 ppm	8. Fixation of CO_2 even less than 10 ppm
9. Less efficient due to higher photorespiration	9. More efficient due to less photorespiration
10. RUBP carboxylase enzyme used for fixation	10. PEP carboxylase and RUBP carboxylase used
11. 18 ATPs used to synthesize one glucose	11. Consumes 30 ATPs to produce one glucose.
12. Example: Paddy, Wheat, Potato and so on	12. Example: Sugar cane, Maize, *Sorghum*, *Amaranthus* and so on

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